# 20261118 Quiz

for (MATH200) Differential Equations

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**Problem 5.2.1.** Consider the differential equation

$$y'' - xy' - y = 0, \qquad x_0 = 1.$$

(a) Seek power series solutions about $x_0$ and find the recurrence relation satisfied by their coefficients.

(b) Find the first four nonzero terms in each of two solutions $y_1$ and $y_2$ (unless the series terminates sooner).

*Solution.* (a) Put $z = x - 1$ and seek a solution in the form

$$y(x) = \sum_{n=0}^{\infty} a_n z^n.$$

Since $x = z + 1$, substituting into $y'' - (z+1)y' - y = 0$ gives

$$\sum_{n=0}^{\infty} (n+1)\left((n+2)a_{n+2} - a_{n+1} - a_n\right) z^n = 0.$$

Thus the recurrence relation is

$$\boxed{a_{n+2} = \frac{a_{n+1} + a_n}{n+2}}, \qquad n \geq 0.$$

(b) Choose $(a_0, a_1) = (1, 0)$ for $y_1$ and $(a_0, a_1) = (0, 1)$ for $y_2$. Applying the recurrence yields

$$\boxed{\begin{aligned} y_1(x) &= 1 + \frac{1}{2}z^2 + \frac{1}{6}z^3 + \frac{1}{6}z^4 + \cdots, \\ y_2(x) &= z + \frac{1}{2}z^2 + \frac{1}{2}z^3 + \frac{1}{4}z^4 + \cdots, \end{aligned}} \qquad z = x - 1.$$

Each series displays its first four nonzero terms.

□

**Problem 5.2.2.** Consider the differential equation

$$y'' - xy' - y = 0, \qquad x_0 = 0.$$

(a) Seek power series solutions about $x_0$ and find the recurrence relation satisfied by their coefficients.

(b) Find the first four nonzero terms in each of two solutions $y_1$ and $y_2$ (unless the series terminates sooner).

(c) By evaluating the Wronskian $W(y_1, y_2)(x_0)$, show that $y_1$ and $y_2$ form a fundamental set of solutions.

(d) If possible, find the general term in each solution.

*Solution.* (a) Substitute $y(x) = \sum_{n=0}^{\infty} a_n x^n$ into the equation:

$$\sum_{n=0}^{\infty} (n+1)\left((n+2)a_{n+2} - a_n\right) x^n = 0.$$

Therefore

$$\boxed{a_{n+2} = \frac{a_n}{n+2}}, \qquad n \geq 0.$$

(b) Choosing $(a_0, a_1) = (1, 0)$ and $(0, 1)$, respectively, gives

$$\boxed{\begin{aligned} y_1(x) &= 1 + \frac{x^2}{2} + \frac{x^4}{8} + \frac{x^6}{48} + \cdots, \\ y_2(x) &= x + \frac{x^3}{3} + \frac{x^5}{15} + \frac{x^7}{105} + \cdots. \end{aligned}}$$

(c) The chosen initial coefficients give $y_1(0) = 1$, $y_1'(0) = 0$, $y_2(0) = 0$, and $y_2'(0) = 1$. Hence

$$\boxed{W(y_1, y_2)(0) = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0}.$$

Thus $y_1$ and $y_2$ are linearly independent and form a fundamental set of solutions.

(d) Iterating the recurrence separately for even and odd indices gives

$$a_{2n} = \frac{a_0}{2^n n!}, \qquad a_{2n+1} = \frac{a_1}{(2n+1)!!}, \qquad n \geq 0,$$

where $(2n+1)!! = 1 \cdot 3 \cdots (2n+1)$. Consequently,

$$\boxed{\begin{aligned} y_1(x) &= \sum_{n=0}^{\infty} \frac{x^{2n}}{2^n n!}, \\ y_2(x) &= \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!!}. \end{aligned}}$$

□

**Problem 5.3.4.** Determine a lower bound for the radius of convergence of series solutions of

$$y'' + 4y' + 6xy = 0$$

about each of the points $x_0 = 0$ and $x_0 = 3$.

*Solution.* In standard form the coefficients $p(x) = 4$ and $q(x) = 6x$ are entire. There are no finite singular points, so the series solutions centered at either ordinary point converge for all real $x$ (indeed, for all complex $x$). Thus

$$\boxed{R = \infty \quad \text{for } x_0 = 0 \text{ and } x_0 = 3}.$$

□

**Problem 5.3.5.** Determine a lower bound for the radius of convergence of series solutions of

$$(x^2 - 2x - 3)y'' + xy' + 4y = 0$$

about $x_0 = 5$, $x_0 = -5$, and $x_0 = 0$.

*Solution.* The coefficients in standard form are

$$p(x) = \frac{x}{(x-3)(x+1)}, \qquad q(x) = \frac{4}{(x-3)(x+1)}.$$

Their singular points are $x = 3$ and $x = -1$. The distance from each center to the nearest singular point gives the guaranteed radius:

$$\begin{aligned} x_0 = 5: &\quad R \geq \min(2, 6) = \boxed{2}, \\ x_0 = -5: &\quad R \geq \min(8, 4) = \boxed{4}, \\ x_0 = 0: &\quad R \geq \min(3, 1) = \boxed{1}. \end{aligned}$$

□

**Problem 5.3.6.** Determine a lower bound for the radius of convergence of series solutions of

$$xy'' + y = 0$$

about $x_0 = 2$.

*Solution.* Dividing by $x$ gives $y'' + y/x = 0$, whose only finite singular point is $x = 0$. Its distance from the center $x_0 = 2$ is 2. Hence the theorem guarantees

$$\boxed{R \geq 2}.$$

□

**Problem 5.3.8.** For the differential equation

$$e^x y'' + xy = 0,$$

find the first four nonzero terms in each of two power series solutions about the origin. Show that they form a fundamental set of solutions. What do you expect the radius of convergence to be for each solution?

*Solution.* Dividing by $e^x$ gives $y'' + xe^{-x}y = 0$. Write

$$y = \sum_{n=0}^{\infty} a_n x^n, \qquad xe^{-x} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(k-1)!} x^k.$$

Equating coefficients yields

$$(n+2)(n+1)a_{n+2} + \sum_{k=1}^{n} \frac{(-1)^{k-1}}{(k-1)!} a_{n-k} = 0, \qquad n \geq 0.$$

Taking $(a_0, a_1) = (1, 0)$ and $(0, 1)$ gives

$$\boxed{\begin{aligned} y_1(x) &= 1 - \frac{x^3}{6} + \frac{x^4}{12} - \frac{x^5}{40} + \cdots, \\ y_2(x) &= x - \frac{x^4}{12} + \frac{x^5}{20} - \frac{x^6}{60} + \cdots. \end{aligned}}$$

Since $W(y_1, y_2)(0) = 1$, these solutions form a fundamental set. Both coefficients in the standard equation, 0 and $xe^{-x}$, are entire, so the radius of convergence of each series is

$$\boxed{R = \infty}.$$

□

**Problem 5.4.1.** Find the general solution of

$$x^2y'' + 5xy' + 3y = 0$$

valid on any interval not containing the singular point.

*Solution.* For the Euler equation, set $y = x^r$ on $x > 0$. The indicial equation is

$$r(r-1) + 5r + 3 = (r+1)(r+3) = 0.$$

The two power functions also solve the equation for $x < 0$, giving

$$\boxed{y(x) = c_1x^{-1} + c_2x^{-3}}, \qquad x \neq 0,$$

with arbitrary constants on each interval not containing 0.

□

**Problem 5.4.3.** Find the general solution of

$$x^2y'' - 5xy' + 9y = 0$$

valid on any interval not containing the singular point.

*Solution.* The indicial equation

$$r(r-1) - 5r + 9 = (r-3)^2 = 0$$

has the repeated root $r = 3$. The Euler equation therefore has the general solution

$$\boxed{y(x) = x^3(c_1 + c_2 \ln|x|)}, \qquad x \neq 0,$$

on any interval not containing 0.

□

**Problem 5.4.4.** Find the general solution of

$$x^2y'' + 3xy' + 10y = 0$$

valid on any interval not containing the singular point.

*Solution.* The indicial equation is

$$r(r-1) + 3r + 10 = r^2 + 2r + 10 = 0,$$

whose roots are $r = -1 \pm 3i$. Therefore

$$\boxed{y(x) = |x|^{-1}\left(c_1\cos(3\ln|x|) + c_2\sin(3\ln|x|)\right)}, \qquad x \neq 0.$$

The constants are chosen independently on each interval not containing 0.

□

**Problem 5.4.8.** Find the general solution of

$$(x-1)^2y'' + 8(x-1)y' + 12y = 0$$

valid on any interval not containing the singular point.

*Solution.* Set $z = x - 1$. The equation becomes $z^2y'' + 8zy' + 12y = 0$, with indicial equation

$$r(r-1) + 8r + 12 = (r+3)(r+4) = 0.$$

Thus

$$\boxed{y(x) = c_1(x-1)^{-3} + c_2(x-1)^{-4}}, \qquad x \neq 1,$$

on any interval not containing 1.

□

**Problem 5.4.11.** Solve the initial value problem

$$x^2y'' - 3xy' + 4y = 0, \qquad y(-1) = 2, \qquad y'(-1) = 2.$$

Plot the graph of the solution and describe its behavior as $x \to 0$.

*Solution.* The indicial equation is

$$r(r-1) - 3r + 4 = (r-2)^2 = 0.$$

As the initial point is $-1$, the relevant interval is $x < 0$, on which

$$y = x^2(c_1 + c_2 \ln(-x)).$$

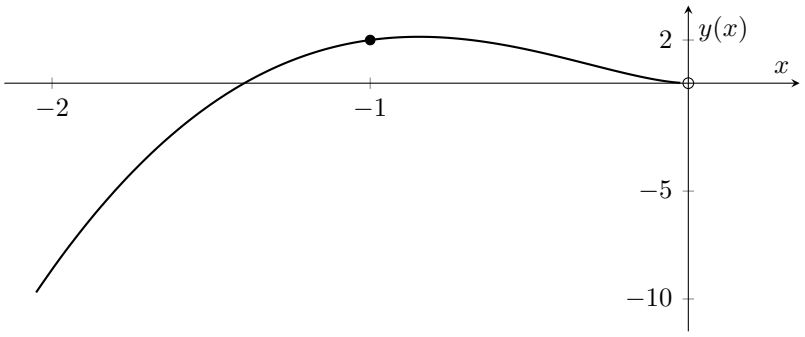
From $y(-1) = c_1 = 2$ and $y'(-1) = -2c_1 - c_2 = 2$, we obtain $c_2 = -6$. Hence

$$\boxed{y(x) = x^2(2 - 6\ln(-x))}, \qquad x < 0.$$

Since $x^2 \ln(-x) \to 0$ as $x \to 0^-$, the solution approaches zero:

$$\boxed{\lim_{x \to 0^-} y(x) = 0}.$$

The graph is shown below, with an open circle at $x = 0$ because the differential equation is singular there.




□